

Deep research: data quality, strategy quality, and the futures question

Generated 2026-05-12. Source: full audit of `services/ingest/ingest/collectors/`, `user_data/strategies/SentimentOnchainStrategy.py`, and live Postgres queries against the running stack.

Methodology

Every claim about the bot is grounded in a file I read or a query I ran. Claims about general trading practice based on training (not measurement) are marked **[opinion]**. No fabricated numbers.

PART 1 — Data: what the bot actually sees today

1.1 Per-coin metrics flowing in (last 24h, verified)

Metric	Rows / 24h
price_usd	93,976
volume_24h	93,975
price_change_24h	93,974
funding_rate	79,525
mark_price	79,525
open_interest	77,950
top_trader_lsr	8,098
global_account_lsr	8,095
taker_buy_sell_ratio	7,935

chain_tvl	1,232
fear_greed_index	1

Source breakdown by file:

- `binance_spot.py` → `/api/v3/ticker/24hr` once per cycle → `price_usd`, `volume_24h`, `price_change_24h`
- `binance_derivatives.py` → `/fapi/v1/premiumIndex` + per-symbol `/fapi/v1/openInterest` → `funding_rate`, `mark_price`, `open_interest`
- `binance_ratios.py` → three endpoints (`topLongShortAccountRatio`, `globalLongShortAccountRatio`, `takerLongShortRatio`)
- `defillama.py` → `/v2/chains` → `chain_tvl` for 18 chains
- `fear_greed.py` → `alternative.me` → `fear_greed_index`

1.2 Text / sentiment sources (verified)

- `news_rss.py` — six RSS feeds: CoinDesk, CoinTelegraph, Decrypt, BitcoinMagazine, TheBlock, BeInCrypto
- `reddit_rss.py` — public RSS from configured subreddits (`cryptocurrency`, `bitcoin`, `cryptomarkets`)
- `reddit.py` — `asyncpraw`, only runs if credentials present (currently disabled)
- `cryptocompare_news.py` — public news endpoint
- `cryptopanic.py` — only runs if token set (currently disabled)
- `lunarcrush.py` — only runs if token set (currently disabled)
- `telegram_channels.py` — seven public channels: CoinDesk, WhaleAlert, CryptoCurrencyNews, CoinTelegraph, `intothecryptoverse_news`, CoinMarketCap, `SplexCrypto_News`

All text flows into `posts`, CryptoBERT scores it into `sentiment_scores`, the aggregator rolls those into 1h `sentiment_aggregates`.

1.3 What the strategy actually consumes

`SentimentOnchainStrategy.py` reads these per-row features:

From OHLCV (delivered by freqtrade/CCXT, not by ingest): close, high, low, open, volume.

From Postgres aggregates:

- sentiment_mean, sentiment_z, sentiment_count
- funding_rate, funding_available
- oi_z
- tvl_z, has_tvl_data
- fng, fng_available
- btc_trend_up (computed from BTC 1h OHLCV)

Critical finding: Of the eleven metric types collected, the strategy uses only five. The rest is dead data:

Collected	Used by strategy?
price_usd (Binance ticker)	No — strategy uses OHLCV from CCXT instead
volume_24h	No
price_change_24h	No — strategy computes its own 1h change
mark_price	No
top_trader_lsr	Partial — only in <code>_momentum_veto_reason</code> (soft veto), never in entry rules
global_account_lsr	No — never queried
taker_buy_sell_ratio	No — never queried

That is **three valuable signals (top-trader LSR, retail LSR, taker buy/sell ratio) being collected at roughly 24,000 rows per day and ignored.**

1.4 What's NOT collected that crypto traders typically use

These are things I checked and confirmed are not collected. **[opinion]** on value:

Data	Cost	Value
Order-book depth (bids/asks within 1% of mid)	Free (Binance REST)	High. Detects whale walls and absorption. Single biggest missing piece for short-term entries.

Liquidations data (Binance forced-close feed)	Free (websocket) or Coinglass	High. Big liquidation cascades mark capitulation lows and blow-off tops.
Exchange netflow (coins moving in/out of Binance)	Glassnode paid / Etherscan free	Medium. Inflows often precede selling.
Stablecoin supply / movements	DefiLlama free / Glassnode	Medium. USDT minting often precedes broad rallies.
Whale wallet flows (top-100 holders)	Etherscan / Arkham (paid)	Medium-high.
Token unlock schedules	TokenUnlocks.app free	Medium. Coins about to unlock supply often dump.
DEX vs CEX volume ratio	DefiLlama free	Low-medium. Useful for memes/long-tail.
X/Twitter firehose	Paid (\$100+/mo)	Medium. Mostly noise but does lead price.
Alpha Telegram/Discord groups	Manual curation	Medium. Quality varies wildly.
Real OHLCV stored (multi-timeframe, all pairs)	Free	High. Currently freqtrade re-downloads on every backtest.
Funding rate predictions	Calc from premium index	Medium. Already collected raw; not pre-computed forward.

Conclusion on data quality: The bot has a strong foundation (free APIs, broad coverage, real-time). The most actionable gaps are (1) **using the LSR ratios already being collected**, (2) **adding order-book depth**, and (3) **adding Binance liquidations feed**. Everything else is incremental.

PART 2 — Indicators: what the strategy actually computes

2.1 The complete TA indicator inventory

Every `ta.*` call in the strategy file:

```
ta.RSI(dataframe, timeperiod=14) # 5m RSI ta.EMA(dataframe, timeperiod=12) # 5m
fast EMA ta.EMA(dataframe, timeperiod=50) # 5m slow EMA ta.RSI(informative,
timeperiod=14) # 1h RSI ta.EMA(informative, timeperiod=12) # 1h fast EMA
ta.EMA(informative, timeperiod=50) # 1h slow EMA ta.EMA(btc_1h, timeperiod=48) #
BTC 1h regime fast ta.EMA(btc_1h, timeperiod=200) # BTC 1h regime slow
```

Total: two unique indicators (RSI, EMA). Everything else (pump_size_pct, volume_ratio, pullback_from_high_pct, sentiment_z, oi_z, etc.) is computed by hand from OHLCV or from DB lookups.

2.2 What common technical indicators are missing

[opinion] on each:

Indicator	What it tells you	Likely value here
MACD	Trend reversal momentum	Low-medium — overlaps with EMA crosses
Bollinger Bands	Volatility bands, mean reversion	Medium-high — would identify "pulled back to lower band" entries
ATR (Average True Range)	Volatility	High — could replace fixed 3% stop-loss with volatility-scaled stops
VWAP	Volume-weighted price benchmark	Medium — useful for intraday bias
Stochastic RSI	More sensitive momentum	Low — duplicates RSI signal
OBV (On-Balance Volume)	Volume-flow direction	Medium
Pivot points / S/R levels	Where buyers/sellers act	High — pullback strategies benefit hugely from context
Market structure (HH/HL/LL/LH)	Trend state	High — distinguishes pullback in uptrend from failed bounce in downtrend
Candlestick patterns (engulfing, pin bar)	Reversal signals	Low-medium
Heikin Ashi	Smoothed trend visualization	Low

2.3 What's specifically wrong with the current indicator set

Concrete observations:

Fixed 3% stop-loss for everything (`stoploss = -0.03`). A 3% stop on BTC (daily ATR ~2%) is reasonable. A 3% stop on a meme coin (daily ATR ~10%) is noise — you'll get stopped out on normal wiggle. ATR-based stops would help materially **[opinion]**.

EMA(12)/EMA(50) on 5m is sensitive to noise. Whipsaw is constant. Most pro setups use EMA(20)/EMA(50) or EMA(21)/EMA(55) **[opinion]**.

The "1h informative" timeframe is the only multi-timeframe context. No daily or weekly trend. The BTC regime gate is the first piece of cross-pair context the strategy has, added today.

No price-relative-to-level features. "How far from yesterday's high?" or "is current price above the prior weekly close?" — simple, free, and absent.

2.4 The "rule-based vs ML" question

The bot is 100% rule-based. CryptoBERT scores sentiment but does not predict price. **[opinion]**: ML for crypto price prediction is hard and overrated — most retail-built ML bots fail because the signal-to-noise ratio is low and the data quality bar is high. The features the strategy already computes are exactly the kind of inputs that would feed a lightweight model (e.g. LightGBM). But you should not put ML on top of an unproven strategy — first prove the rules work, then add ML to extract more juice. Doing it the other way is "we don't know which inputs matter, let's hope the algorithm figures it out", and it doesn't.

PART 3 — Strategy quality

3.1 What kind of strategy is this?

After v2 changes:

- **Family**: Momentum-pullback ("buy the dip after a breakout")
- **Timeframe**: 5m trades, average hold 18 minutes in backtest
- **Edge hypothesis**: Coins that pump and then briefly retrace tend to continue if the pullback is shallow
- **Regime gate**: Don't trade in a BTC downtrend (medium-term)

This is a reasonable, well-understood strategy family. Many professional discretionary traders use it. It's not exotic, which is good — exotic strategies typically don't survive after fees.

3.2 What measured v2 actually does

From the 30-day backtest:

5 trades | 3W / 2L | 60% win rate Net P&L: +\$0.35 on \$10,000 (0.003%) Best: CHIP/USDT +3.12% Worst: SAGA/USDT -3.01% Profit factor: ~1.05 Drawdown: 0.05%

Measured edge: ~+0.05% per trade. That's the actual statistical edge over 30 days.

3.3 The "is this the best strategy?" question, honestly

[opinion] Short answer: it's a reasonable strategy family but currently a fragile prototype.

Strengths:

- Multi-source feature set (price + sentiment + on-chain + regime)
- Pullback entry is statistically more robust than chasing breakouts
- Regime gate prevents bear-market disasters
- Position-sizing discipline (fixed \$500 / 3 max)

Weaknesses:

- Trade frequency too low — 5 trades / 30 days won't compound
- Win:loss ratio less than 1 — avg winner is smaller than avg loser
- One-size-fits-all stop-loss — 3% on every coin regardless of volatility
- Strict path is dead — 0 trades fired over a month
- No daily/weekly context — operates entirely on 5m + 1h
- Single direction — long-only, can't profit from down-moves

3.4 Alternative strategy families worth knowing about

[opinion] — retail crypto strategy taxonomy:

1. **Trend-following on higher timeframes (1h–4h)** — buy higher highs in confirmed uptrend, exit on trend break. Long holds. Simple. Robust. Few-times-a-month with big winners.
2. **Mean reversion in range markets** — buy oversold (RSI<20, lower Bollinger), sell overbought. Works in chop, blows up in trending moves. Needs regime filter.
3. **Pullback in trend (this bot)** — buy dips in confirmed uptrend. Medium frequency.
4. **Breakout** — buy when price crosses a multi-day high. High false-positive rate; needs strong filters.

5. **Cross-exchange arbitrage** — capture price differences between Binance and other venues.
Capital-intensive but low-risk.
6. **Funding rate arbitrage (futures-only)** — short the perp when funding is extreme positive, hold spot.
Market-neutral. Real edge but small per trade.
7. **Event/news trading** — fire on specific catalysts. The catalyst-pairlist work is the foundation for this.

The current pullback strategy is in family #3. The closest pivot that might add edge without changing the family is to **also** look for **breakouts of consolidation ranges** (family #4) when the regime is bullish.

PART 4 — Futures and shorts: the honest economic analysis

4.1 Cost structure differences (current Binance rates)

Cost	Binance Spot	Binance USD Futures
Maker fee	0.1% (0.075% w/ BNB)	0.02% (0.018% w/ BNB)
Taker fee	0.1% (0.075% w/ BNB)	0.04% (0.036% w/ BNB)
Funding rate	N/A	Paid every 8h, range: typically -0.075% to +0.1% per period
Borrow cost	N/A	Built into funding

Futures fees are actually **lower** than spot. The friction is the **funding rate**, paid every 8 hours by longs (when positive) or shorts (when negative). In a strong bull market, longs can pay 0.05–0.1% per 8h = 0.15–0.3% per day just to hold. For a strategy holding 18 minutes on average, funding is irrelevant — you never see a funding event.

4.2 The killer math for the current strategy on futures

Round-trip cost of one trade:

- Spot taker: $0.075\% \times 2 = \mathbf{0.15\% \text{ per trade}}$
- Futures taker: $0.036\% \times 2 = \mathbf{0.072\% \text{ per trade}}$

- Plus typical slippage: ~0.05% **[opinion]**

Total round-trip friction:

- Spot: ~0.20%
- Futures: ~0.12%

Current strategy's average winning trade: ~1.04%. Average losing trade: ~-2.02%. Net per trade: **+0.05%**.

Spot	Futures (1×)	Futures (3×)	Futures (10×)
Strategy edge per trade	+0.05%	+0.15% on stake	+0.50% on stake
Round-trip cost	-0.12%	-0.36%	-1.20%
Net per trade	-0.07%	-0.21%	-0.70%

Critical finding: the current strategy's edge is below the round-trip cost on spot — meaning the "\$0.35 over 30 days" backtest result assumed paper-trading without realistic slippage. In the real world the strategy may already be slightly negative.

Switching to futures at any leverage with the current strategy = guaranteed loss.

4.3 Liquidation risk at common leverage levels

Approximation (ignoring small maintenance-margin adjustments):

- 1× leverage: no liquidation risk (same as spot)
- 2× leverage: liquidated at ~50% adverse move
- 3× leverage: liquidated at ~33% adverse move
- 5× leverage: liquidated at ~20% adverse move
- 10× leverage: liquidated at ~10% adverse move
- 20× leverage: liquidated at ~5% adverse move

[opinion] Crypto sees 10–20% single-day moves routinely. Luna: -99% in 3 days. FTX: -90% in 48h. Anything above 3× leverage on a single position is gambling, not trading. For a momentum strategy with 3% hard stops, 2–3× leverage is the practical max — and even that requires the stop-loss to never slip during a fast move.

4.4 Do shorts work in crypto historically?

[opinion]:

- **Bull markets:** shorts are graveyards. Coins that "look done" can melt up another 100%.
- **Bear markets:** shorts work, but the easy short trades happen in the first 30% of the drawdown — by the time most people recognize "we're in a bear", the easy money is gone.
- **Ranging markets:** short the resistance, long the support. Mean-reversion shorts. Bot-friendly.
- **Single-event shorts:** token unlocks, security exploits, exchange depegs. High edge if you're early; impossible to scale because they're rare.
- **Funding-rate shorts:** when perp funding is extreme positive, short and earn the funding while you're correct on direction. Real market-neutral edge.

The bot's current regime gate gives a partial answer to "long or short": when $\text{BTC EMA48} < \text{EMA200}$, don't long. The mirror would be: when $\text{BTC EMA48} < \text{EMA200}$, consider shorting bounces in weak coins.

4.5 Is shorting worth it for this bot, today?

Direct answer: not yet, and putting the current strategy on futures of any flavor will lose money.

Three conditions need to be met first:

The long-side strategy needs to clear round-trip fees on average. It doesn't. Win:loss ratio improvement is required. Either tighter stops, partial-profit-taking, or wider thresholds for catching bigger moves.

The bot needs a separate short-trigger signal distinct from the long trigger. The mirror image of "pullback after pump" isn't "bounce after dump" — bounces after dumps usually succeed in crypto. The real short signal is "failed bounce after dump" — price tries to recover, fails to clear resistance, rolls over. Different pattern.

Sample size on shorts needs to be backtestable. That means at least one bear or distribution market in the backtest window. Our 30-day window has BTC up 26% — there is no shortable regime, so we cannot validate shorts on this data.

To get from here to "futures with shorts" responsibly:

- Tune the long side so it clears futures friction. Bar: avg win $\geq 0.5\%$, win rate $\geq 55\%$, profit factor ≥ 1.4 .

- Wait for or download a bear-market window in the historical data.
- Implement short logic (mirror image with the right pattern) and backtest it on that bear window.
- Only then flip the live config to futures, dry-run for ≥ 2 weeks, and finally consider live.

This sequence is the difference between "futures bot that loses money quickly" and "futures bot that captures both directions of crypto's volatility."

PART 5 — Recommendations, ranked by leverage

Tier A — Cheap, measurable wins (do these next)

Use the unused LSR / taker-buy-sell data. Already in Postgres. Adding `top_trader_lsr_z` and `taker_buy_sell_ratio` as gates is a 30-line change.

Replace the fixed 3% stop with ATR-based stops. `ta.ATR(dataframe, timeperiod=14)` then `stoploss_pct = max(0.015, atr / close * 1.5)`. Volatile coins get wider stops, stable coins get tighter ones.

Add take-profit ladder tuned for measured trade lengths. Backtest showed average winners last 23 min and gain ~1%. Set `minimal_roi` to take partial profit at 0.8% after 15 min, 1.5% after 30 min. Locks in winners.

Loosen the pullback thresholds. Current `pump_size > 8%`, `pullback 2-5%`. Try `pump_size > 5%`, `pullback 1-7%`. Backtest delta tells you if more trades = more profit.

Tier B — Real but bigger work

Add Binance liquidations data. Free WebSocket. Liquidation cascades are clean signals of capitulation lows.

Add order-book depth snapshots. Detects walls and absorption. Free.

Add ATR and Bollinger Bands. Two more talib indicators, instant compute.

Build market-structure detection (HH/HL via pivot highs/lows).

Tier C — Strategy expansion

Add a mean-reversion entry path (oversold bounce) for ranging regimes.

Implement short logic mirroring the long side with the correct pattern (failed bounce, not bounce).
Backtest on bear windows.

Tier D — Only after A–C are validated

Switch to futures in dry-run. Validate that fees + funding don't kill the edge.

ML on top of validated features. LightGBM on the existing feature set, trained to predict next-N-candle return.

Consider paid data (Glassnode, CryptoQuant) only if Tier A–C have diminishing returns.

TL;DR — direct answers

Is the bot getting the best information?

No. It gets good, free, broad information, but with clear gaps: (a) three valuable signals (LSR ratios, taker buy/sell ratio) collected and not used in the strategy; (b) missing order-book depth and liquidations, both free and high-value; (c) missing common indicators like ATR and Bollinger.

Is the strategy the best?

Reasonable family (pullback-in-trend), but currently a thin prototype: 5 trades / 30 days, +0.05% per trade, **likely below round-trip fees in real conditions**. Structural fixes (ATR stops, ROI ladder tuning, more triggers) are clear next steps.

Is futures + shorts worth it?

Not yet. Current edge is below the fee floor on spot already, futures funding cost would push it deeper into the red, and shorts can't be validated against our current (all-bull-market) backtest data. Honest sequence: (1) tune longs to clearly clear fees, (2) acquire bear-market historical data, (3) write and backtest short logic, (4) switch live to futures dry-run, (5) only then real money on shorts. Anything faster is gambling, not trading.